

Mathematical Statistics II

Maximum Likelihood Estimation

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Introduction

Overview

- The next approach we will discuss is Maximum Likelihood Estimation (MLE).
- As we will see, the MLE has several desirable properties, and as a result is often favored over approaches like the method of moments.
- The material for this section largely comes from Chapter 8.5 of Rice (2007), and various sections in Pawitan (2001).

maximum likelihood :
estimation
→ estimator
→ estimate.

} MLE

$$\hat{\theta} = f(x_i)$$

→ estimate.

Likelihood: an introduction

What is likelihood?

- The term “likelihood” is often used colloquially to mean something analogous to probability. E.g., “What is the likelihood that it rains tomorrow?”
- When we use this term in statistics / mathematics, we mean something specific that isn’t the same thing as probability.
- The use of the term “likelihood” was first made by R. A. Fisher, who was the architect and primary proponent of “likelihood-based-inference”.
- We will start with the treatment of likelihood in the text “In all Likelihood” (Pawitan, 2001), which is a fantastic resource on the subject. (This will lead to some review...)

What is likelihood? II

Coin Flips

We will revisit this example, as it is a great starting point to connect with existing understanding.

Consider flipping a coin $N = 10$ times.

X is # heads, $N = \text{totals}$, p is prob of heads -

$$\rightarrow X \sim \text{Bin}(N, p).$$

Suppose $p = 0.4$, What is the prob. of observing $X=8$?

$$P_{\substack{\theta=p \\ =0.4}}(X=8) = 0.011.$$

• If we flip 10 coins, 10,000 times, we would expect to see $X=8$ about $(0.011)(10,000) = \underline{\underline{110}}$ times.

- Suppose p is unknown,
we observe $X=8$.

- $p \neq 0$, $p \neq 1$

- $p \approx 0$, "unlikely"

- suppose $p = 0.1$ $\rightarrow P_{0.1}(X=8) = 3.6 \times 10^{-7}$

• $p \leftarrow$ using X alone.

$\rightarrow$ no priors.

• $p = 0.6$?

$$p_{p=0.6}(X=8) = 0.12$$

What is likelihood? III

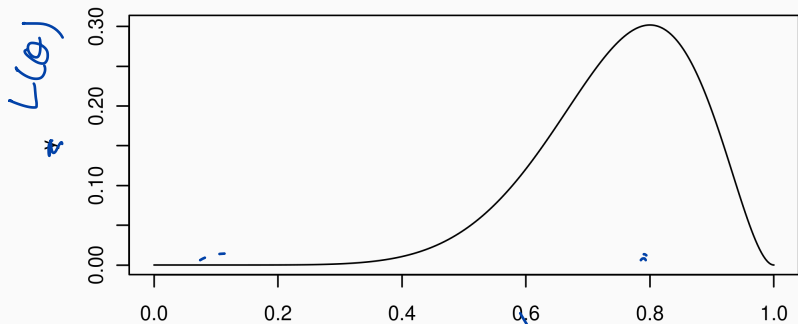
- For our specific coin-flipping example with $N = 10$, $X = 8$, the likelihood function is

$$L(\theta) = P_{\theta}(X = 8).$$

- This is plotted in the following way:

```
theta  
X <- seq(1e-8, 1-1e-8, length.out = 1000)  
L_theta <- dbinom(8, 10, X)  
plot(x = X, y = L_theta, type = 'l')
```

What is likelihood? IV



What is likelihood? V

- From the figure, we see that p is unlikely to be less than 0.5, or greater than 0.95.
- Given the data alone (no prior), we should prefer a value somewhere in the middle of these values.
- We still have some uncertainty about the value of p , but the likelihood gives us a numerical way to compare values of θ . Stochastic uncertainty as a result of sampling is captured in the likelihood function $L(\theta)$.

What is likelihood? VI

- The likelihood is not a probability. Though it came from a probability, the likelihood function (a function of θ) does not satisfy the requirements to be a probability. In our previous example, we have: $L(\theta) = P_{\theta}(X=8) = \binom{10}{8} \theta^8 (1-\theta)^{10-8}$

$$\int_0^1 L(\theta) d\theta = 1/11 \neq 1.$$

- For discrete probability, the likelihood was continuous. Discrete likelihoods are possible, arising when we want to select from a list $\{\theta_1, \theta_2, \dots\}$.



What is likelihood? VII

- The idea behind maximum likelihood estimation (MLE) is simple: our estimate is the value of θ that maximizes the likelihood function $L(\theta)$.
- The MLE is considered a *frequentist* approach. Why? It quantifies a maximum belief about a parameter, which is more Bayesian in nature than Frequentist.
- As we'll see later, the MLE has nice theoretical Frequentist *properties*, and as a result can be justified via the frequentist paradigm.
- Still, it has close connection to Bayesian estimation and interpretation. In fact, we'll discuss connections between the MLE and Bayesian statistics later.

What is likelihood? VIII

- Often, maximizing the likelihood directly is challenging, so we maximize the log-likelihood instead.
- Other times, the likelihood has to be maximized numerically.

MLE of coin toss problem

Suppose we have $N = 10$ total tosses, and n total heads. Find the MLE of p , the probability of heads.

$$\begin{aligned} L(\theta) &= P_{\theta=p}(X=n) & x^* &= n \\ &= \binom{N}{n} p^n (1-p)^{N-n} & \text{ell} \\ \ell(\theta) &= \log_e L(\theta) = \cancel{\log \binom{N}{n}} + n \log p + (N-n) \log (1-p) & \text{(ln)} \end{aligned}$$

$$\frac{\partial l}{\partial \theta} = \frac{n}{p} + \frac{N-n}{1-p} (-1)$$

$$= \frac{n}{p} - \frac{N-n}{1-p} = 0$$

$$\frac{n}{p} = \frac{N-n}{1-p} \Rightarrow$$

$$(1-p)n = pN - pn$$

$$n - np = Np - np$$

$$\hat{p} = \frac{n}{N}$$

MoM

$$\longrightarrow = \frac{1}{N} \sum_{i=1}^N X_i$$

Continuous models

$$L(\theta) = P_{\theta}(X = x^*)$$

- The interpretation of the likelihood function as the “the probability of the observed data x^* , considered as a function of θ ” makes perfect sense in the discrete model case.
- For continuous models, the technical issue arises that the probability of any point value x is zero. $L(\theta) = P_{\theta}(X = x) = 0$
- We resolve the problem similar to what was done in Math 4450 and the John Rice text: approximate the probability by discretizing into small, discrete intervals:

$$x^* \in (x^* - \epsilon/2, x^* + \epsilon/2),$$

$$P_{\theta}(X \in (x^* - \epsilon/2, x^* + \epsilon/2))$$

Continuous models II

thus, the probability of observing something ϵ -close to the data is:

$$\begin{aligned} L(\theta) &= P_{\theta}(X \in (x^* - \epsilon/2, x^* + \epsilon/2)) \\ &= \int_{x^* - \epsilon/2}^{x^* + \epsilon/2} f(x; \theta) \, dx \approx \epsilon f(x^*; \theta). \end{aligned}$$

- Then, since the likelihood is only meaningful up to a constant (we will discuss likelihood ratios later), then this has the same behavior as $L(\theta) = f(x^*; \theta)$.
- There are more advanced approaches to this problem, but this simple argument justifies the use of the pdf of a continuous random variable as the likelihood $L(\theta)$.

Continuous models III

- **Going forward:** we once again will generalize a model $f(x; \theta)$ to mean either the pmf or pdf of a random variable. I will often say “density” as a blanket term, even if this corresponds to a pmf, not a density.
- Further, when we “integrate” a density, this means either:

$$\int f(x; \theta) dx, \quad \text{If continuous}$$

or

$$\sum_x f(x; \theta), \quad \text{If discrete.}$$

Joint Probabilities

Likelihood with multiple observations

- Often the data we observe is multi-dimensional, rather than summarized as a single observation.
- In this case, the likelihood θ is still determined via the joint model:

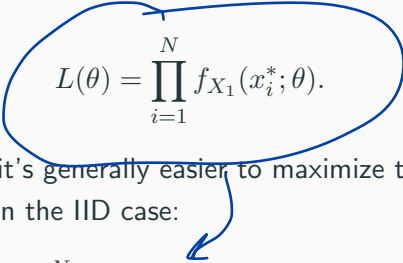
$$L(\theta) = f(x^*; \theta) = f_{X_{1:N}}(x_1^*, x_2^*, \dots, x_N^*; \theta).$$

- We are mostly focused in this class in the case where the observations are independent, meaning the likelihood factors:

$$L(\theta) = \prod_{i=1}^N f_{X_i}(x_i^*; \theta).$$

Likelihood with multiple observations II

- We often further simplify this by assuming the data are identically distributed:


$$L(\theta) = \prod_{i=1}^N f_{X_1}(x_i^*; \theta).$$

- As we've seen, it's generally easier to maximize the log-likelihood. In the IID case:

$$\ell(\theta) = \log \prod_{i=1}^N f_{X_1}(x_i^*; \theta) = \sum_{i=1}^n \log f_{X_1}(x_i^*; \theta).$$

Examples

Examples of finding the MLE

Traffic data: Poisson Model

Returning to a motivating example, suppose we model traffic accidents in a given week as $X_1, X_2, \dots, X_N$, where the data are iid $\text{Poisson}(\lambda)$. Obtain the MLE for λ .

$$f_i(x_i; \lambda) = P(X_i = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$L(\theta) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i^*}}{x_i^*!}$$

$$\ell(\theta) = \sum_{i=1}^n \log \left(\frac{e^{-\lambda} \lambda^{x_i^*}}{x_i^*!} \right)$$

$$= \sum_{i=1}^n \log(e^{-\lambda} \lambda^{x_i^*}) - \log(x^*!)$$

$$= \sum_{i=1}^n \log(e^{-\lambda}) + \log(\lambda^{x_i^*}) - \log(x^*!)$$

$$= \sum_{i=1}^n (-\lambda) + x_i^* \log(\lambda) - \log(x_i^*!)$$

$$= -\lambda n + \log(\lambda) \left(\sum x_i^* \right) - \sum \log(x_i^*!)$$

$$\frac{\partial \ell}{\partial \lambda} = -n + \frac{\sum x_i^*}{\lambda} = 0$$

$\Rightarrow$

$$n = \frac{\sum x_i^*}{\lambda}$$

λ

$\Rightarrow$

$$\hat{\lambda} = \frac{1}{n} \sum x_i^*$$

$\uparrow$

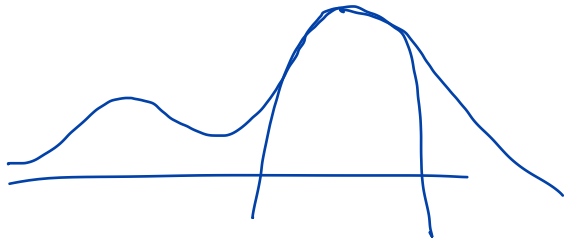
mom.

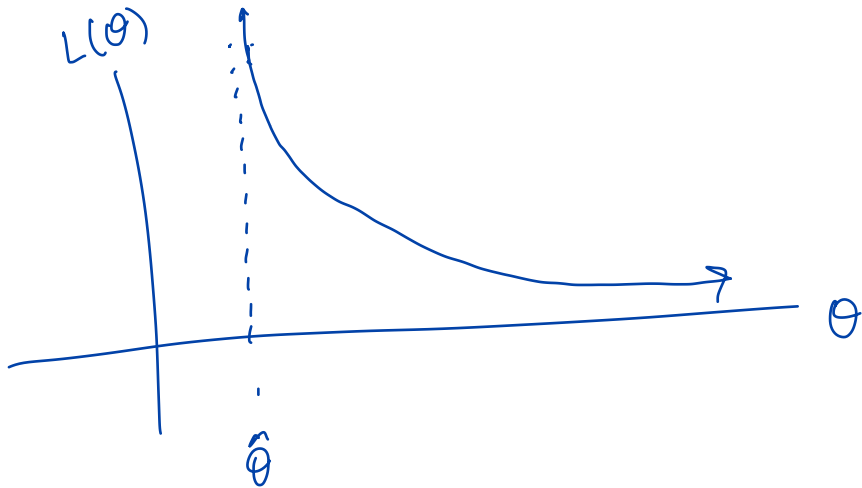
- why do I assume it's a maximum?

- exponential family of distns.

- Normal
- Gamma,
- Beta,
- exponential
- Poisson
- ...

- if n is large, $L(\theta) \rightarrow$ Normal likelihood





Examples of finding the MLE II

Two parameter model: Gaussian model

Suppose we model observations $X_1, \dots, X_N$ as IID $N(\mu, \sigma^2)$ random variables. Find the MLE of $\theta = (\mu, \sigma^2)$.

→ should I find MLE of $\boxed{\sigma}$, or σ^2
MLE is the same for both.

$$X_i \sim \text{iid } N(\mu, \sigma^2)$$

$$\begin{aligned} L(\mu, \sigma) &= \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_i^* - \mu)^2} \\ &= \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i^* - \mu}{\sigma}\right)^2} \end{aligned}$$

$$\ell(\mu, \sigma) = \log \prod_{i=1}^n f_i = \sum \log f_i$$

$$= \sum_{i=1}^n \log \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2} \right)$$

$$= \sum_{i=1}^n \log(1) - \log(\sigma\sqrt{2\pi}) - \frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2$$

$$\underbrace{\left(\log(\sigma) + \frac{1}{2} \log(2\pi) \right)}$$

$$= -n \log \sigma - \frac{n}{2} \log \frac{2}{\pi} - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

$$\frac{\partial \ell}{\partial \mu} = -\frac{1}{2\sigma^2} \frac{\partial}{\partial \mu} \left[\sum_{i=1}^n (x_i - \mu)^2 \right]$$

$$= -\frac{1}{2\sigma^2} \sum_{i=1}^n \frac{\partial}{\partial \mu} (x_i - \mu)^2$$

$$= +\frac{1}{2\sigma^2} \sum_{i=1}^n 2(x_i - \mu) \cdot (+1)$$

$$= \frac{1}{\sigma^2} \sum (x_i - \mu) = \frac{1}{\sigma^2} (\sum x_i - n\mu)$$

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \sigma^{-3} \sum_{i=1}^n (x_i - \mu)^2 \quad \sigma > 0$$

$$= 0 \quad \Rightarrow \quad \frac{n}{\sigma} = \frac{1}{\sigma^3} \sum (x_i - \mu)^2$$

$$\Rightarrow \left(\hat{\sigma}^2 \right) = \frac{1}{n} \sum (x_i - \hat{\mu})^2$$

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum (x_i^* - \mu) = 0$$

$$\Rightarrow \sum (x_i - \mu) = 0$$

$$\sum x_i = n\mu \Rightarrow \mu = \frac{1}{n} \sum x_i$$

$$\mu = \bar{x}$$

~~$$\hat{\theta} = (\bar{x}, \frac{1}{n} \sum (x_i - \bar{x})^2)$$~~

$$\hat{\theta}_{MLE} = (\hat{\mu}, \hat{\sigma}) = (\bar{x}, \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2})$$

$$\hat{\theta}_{MM} = (\bar{x}, \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2})$$

$$X_i \sim N(\mu, \sigma^2)$$

$$\bar{x} = \hat{\mu} \sim N(\mu, \frac{\sigma^2}{n})$$

$$\frac{n\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-1}$$

(Rice, chapter 6)
✓

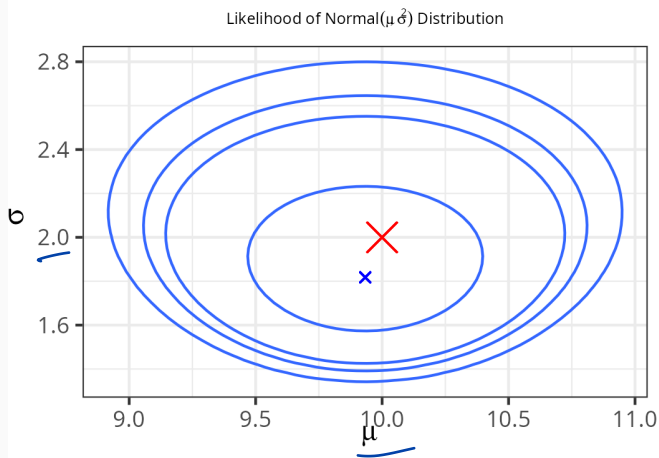


Plotting Normal Likelihood

- The likelihood function (not just to point estimate) will be used to measure uncertainty.
- For models with a single parameter, we often plot the likelihood curve.
- With more than one parameter, however, we have a **likelihood surface**.
- For the iid Normal(μ, σ^2) model, code for plotting this surface is available with course source-code.

```
like <- function(x, theta) {  
  prod(dnorm(x, mean = theta[1], sd = theta  
  sd = theta[2]))  
}
```

Plotting Normal Likelihood II



map2_dbl
↑

Figure 1: Likelihood surface of data generated from normal distribution.

Calculating the likelihood in R

- For standard distributions, the likelihood is easy to calculate in R. Recall the likelihood is:

$$L(\theta) = f(x_i^*; \theta).$$

- For $X_1, \dots, X_n$ iid normal, the likelihood can be calculated using `dnorm`:

$$\theta_{\text{true}} = (4, 3)$$

```
# Synthetic data
x_star <- rnorm(n = 100, mean = 4, sd = 3)
theta <- c(0, 1) # value of theta
Likelihood <- prod(dnorm(x, mean = theta[1], sd = theta[2]))
loglik <- sum(
  dnorm(x, mean = theta[1], sd = theta[2], log = TRUE)
)
```

Numeric Optimization

Numeric Optimization

- In the previous examples, the MLE was available *analytically*.
- In many cases, however, there is no closed-form solution for the MLE, and it must be computed numerically.
- The next example demonstrates this, and then we will discuss optimization strategies.

Numeric Optimization II

Example: Gamma likelihood

Suppose we want to model data $X_1, X_2, \dots, X_n$ as iid $\text{Gamma}(\alpha, \lambda)$, which has the density function:

$$f(x; \alpha, \lambda) = \frac{1}{\Gamma(\alpha)} \lambda^\alpha x^{\alpha-1} e^{-\lambda x}, \quad 0 \leq x < \infty.$$

Find the MLE of $\theta = (\alpha, \lambda)$.

$$L(\theta) = \prod_{i=1}^n f_i(x_i^*; \alpha, \lambda)$$

$$= \prod_{i=1}^n \frac{1}{\Gamma(\alpha)} \lambda^\alpha (x_i^*)^{\alpha-1} e^{-\lambda x_i^*}$$

$$\ell(\alpha, \lambda) = \sum_{i=1}^n \alpha \log \lambda + (\alpha-1) \log x_i^* - \lambda x_i^* - \log \Gamma(\alpha)$$

$$= n\alpha \log \lambda + (\alpha-1) \sum \log x_i^* - \lambda \sum x_i^* - n \log \Gamma(\alpha)$$

$$\begin{aligned} \frac{d\ell}{d\alpha} &= n \log \lambda + \sum \log x_i^* - n \frac{1}{\Gamma(\alpha)} \cdot \Gamma'(\alpha) \\ &= n \log \lambda + \sum \log x_i^* - n \left(\frac{\Gamma'(\alpha)}{\Gamma(\alpha)} \right) = 0 \end{aligned}$$

$$\frac{d\ell}{d\lambda} = \frac{n\alpha}{\lambda} - \sum x_i^* = 0$$

digamma

$$\frac{\partial \ell}{\partial \lambda} = 0 \Rightarrow \frac{n\alpha}{\lambda} = \sum x_i^*$$

$$\Rightarrow \frac{\alpha}{\lambda} = \overline{x_i^*}$$

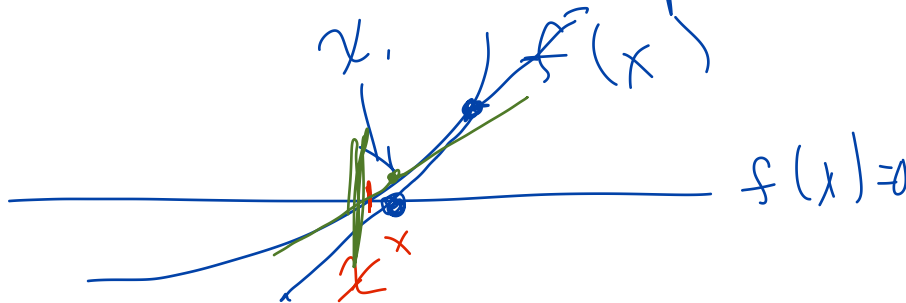
$$\Rightarrow \hat{\lambda} = \frac{\hat{\alpha}}{\overline{x_i^*}}$$

$$\begin{aligned} \frac{\partial \ell}{\partial \alpha} &= n \log \lambda + \sum \log x_i^* - n \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} = 0 \\ &= n \log \frac{\hat{\alpha}}{\overline{x_i^*}} + \sum \log x_i^* - n \frac{\Gamma'(\hat{\alpha})}{\Gamma(\hat{\alpha})} = 0 \end{aligned}$$

↑

$$f(x) = 0$$

Newton-Raphson



x_0

Numeric Optimization III

- The previous example leads us to consider numeric techniques for optimization and root finding.
- Note that this class is **not** an optimization course, so we'll only cover some of the most basic ideas.
- More modern and efficient numeric optimization techniques are readily available in R (or any other statistical software).
- For this class, we'll introduce some basic ideas like the Newton-Raphson approach for root finding and optimization, as well as some other basic methods.

Newton-Raphson root-finding algorithm

- Idea: start at a point θ_0 , and approximate find the tangent line of $f(\theta)$ at the point θ_0 :

$$y - f(\theta_0) = f'(\theta_0)(\theta - \theta_0)$$

- Then, find the root of the tangent line by setting $y = 0$, and solving for θ :

$$\theta = \theta_0 - \frac{f(\theta_0)}{f'(\theta_0)}.$$

Newton-Raphsom root-finding algorithm II

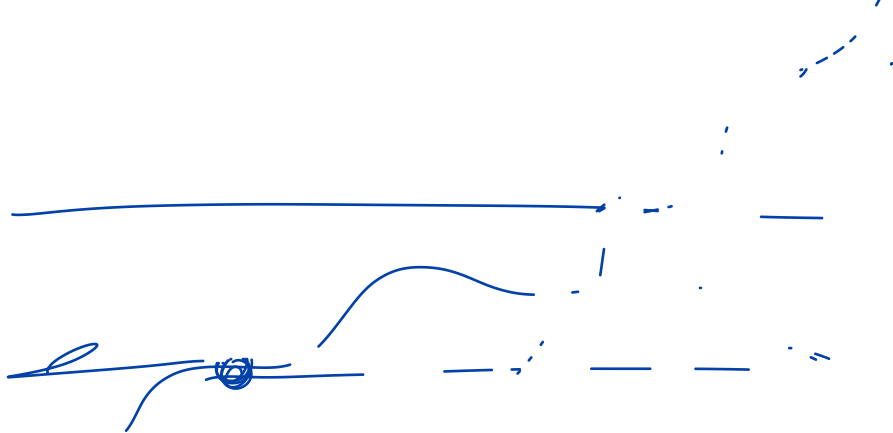
- This root of the tangent line will be closer than our original guess θ_0 , so we set:

$$\theta_1 = \theta_0 - \frac{f(\theta_0)}{f'(\theta_0)},$$

and repeat:

$$\underline{\theta_{n+1}} = \underline{\theta_n} - \frac{f(\theta_n)}{f'(\theta_n)}.$$

- We stop based on some convergence criteria, often something like $|\theta_{n+1} - \theta_n| < \epsilon$, for a small choice of ϵ .
- (In class, check out wikipedia or draw a picture).



Newton-Raphsom root-finding algorithm III

- We need now a starting point θ_0 .
- Really we can pick anything, but it's best if we are close to the MLE.
- For our current problem (Gamma distribution), we could use the MoM estimator:

$$\hat{\alpha}_{\text{MoM}} = \theta_0 = \frac{(\bar{x}_n^*)^2}{\frac{1}{n} \sum_{i=1}^n (x_i^* - \bar{x}_n^*)^2}.$$

$$= n \log \frac{\alpha}{\bar{x}_i^*} + \sum \log x_i^* - n \frac{\Gamma'(\hat{\alpha})}{\Gamma(\alpha)} = 0$$

Newton-Raphsom root-finding algorithm IV

```
NR_root <- function(theta0, fn, deriv, tol = 1e-8, maxiter = 100) {  
  iter <- 0  
  theta_old <- theta0  
  theta_new <- theta_old + 10 * tol  
  
  while(abs(theta_old - theta_new) > tol && iter < maxiter) {  
    iter <- iter + 1  
    theta_old <- theta_new  
    theta_new <- theta_old - fn(theta_old) / deriv(theta_old)  
  }  
  cat("iters: ", iter, "\n")  
  theta_new  
}
```

$$= n \log \frac{\alpha}{\bar{x}_i} + \sum \log x_i - n \frac{\Gamma'(\hat{\alpha})}{\Gamma(\hat{\alpha})} = 0$$

Newton-Raphsom root-finding algorithm V

```
alpha_fn <- function(alpha, data) {  
  n <- length(data)  
  n * log(alpha) - n * log(mean(data)) + sum(log(data)) - n * di  
}  
  
alpha_deriv <- function(alpha, data) {  
  n <- length(data)  
  (n/alpha) - n * psigamma(alpha, 1)  
}  
  
set.seed(123)  
data <- rgamma(n = 23, shape = 1, rate = 2)
```

$-n \times \text{digamma}$

Newton-Raphsom root-finding algorithm VI

Not the exact MoM estimate, but close enough:

```
alpha_mom <- (mean(data)^2) / sd(data)
```

```
NR_root(  
  theta0 = alpha_mom,  
  fn = function(x) alpha_fn(x, data = data),  
  deriv = function(x) alpha_deriv(x, data = data),  
  tol = 1e-10  
)
```

```
iters: 7
```

```
[1] 0.9728019
```

Newton-Raphson root-finding algorithm VII

- Once the estimate of α is found, we can then plug it in to get the estimate of λ :

$$\hat{\lambda} = \frac{\hat{\alpha}}{\bar{x}_n^*} = 1.981.$$

Newton-Raphson root-finding algorithm VIII

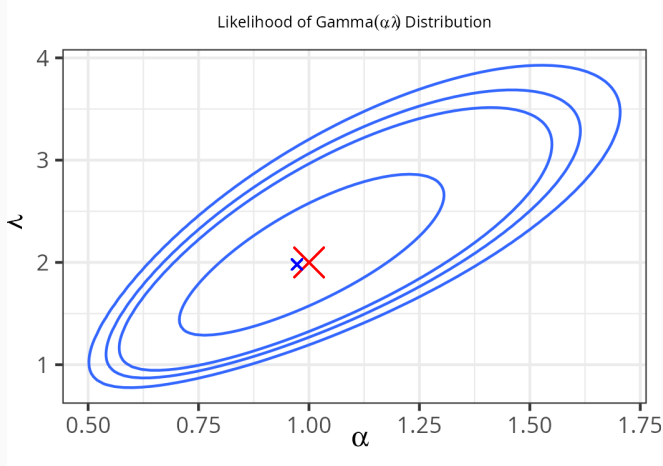


Figure 2: Likelihood surface of data generated from Gamma distribution.

$$\hat{\lambda} = \frac{\sum_{i=1}^n x_i}{n}$$

Root-find considerations

- The function that we built for solving $f(\theta) = 0$ requires the derivative $f'(\theta)$.
- In some cases, the derivative is not readily available. Instead, we can approximate using the definition:

$$f'(\theta) = \lim_{h \rightarrow 0} \frac{f(\theta + h) - f(\theta)}{h} \approx \frac{f(\theta + \Delta) - f(\theta)}{\Delta}.$$

- We just pick Δ to be small, and we get a very good approximation of the derivative.
- Thus, we don't really need to derivative for uni-variate root-finding.
- The same mathematical approach can readily be extended into higher dimensional θ , replacing the derivative with a **gradient**.

Root-find considerations II

- An important consideration, however, is that the results may depend on your starting parameter θ_0 . In practice, you may want to try multiple values of θ_0 .

Root-find considerations III

- In most programming languages, there will be pre-built methods for root-finding and optimization.
- Our function we built works, but it doesn't do careful error checking, and it isn't optimized for speed.
- For univariate-root finding $f(\theta) = 0$ in $\mathbb{R}$, we can use the `uniroot` function, which doesn't require a derivative.
- This function is very efficient for solving roots if $\theta \in \mathbb{R}$. For higher dimensions, we need to import a different package.

→ $f_n(x)$ has to take 1 arg

→ interval for x .

Root-find considerations IV

```
uniroot(  
  function(x) alpha_fn(x, data = data),  
  interval = c(0.5, 2)  
)
```

\$root

[1] 0.9728068

\$f.root

[1] -7.754612e-05

\$iter

[1] 6

\$init.it

[1] NA

Handwritten notes: $\rightarrow X$ (pointing to the closing parenthesis), $\leftarrow X \$ root +$ (pointing to the root value), and a circled $\$$.

Feb 10 : Review

l(θ) = ^{i.i.d.} $\prod_{i=1}^n \underline{f(x_i; \theta)}$ ← pdf, pmf

$$l(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$$

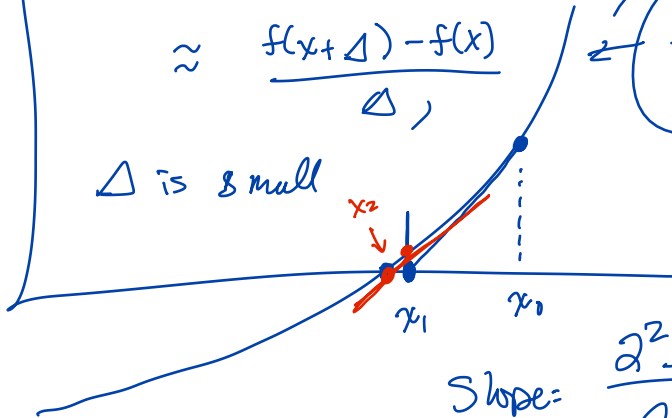
$$\left| \frac{\partial l}{\partial \theta} = 0 \Rightarrow \hat{\theta}_{MLE} \right|$$

$$f'(x) = \lim_{\Delta \rightarrow 0} \frac{f(x+\Delta) - f(x)}{\Delta}$$

$$\approx \frac{f(x+\Delta) - f(x)}{\Delta},$$

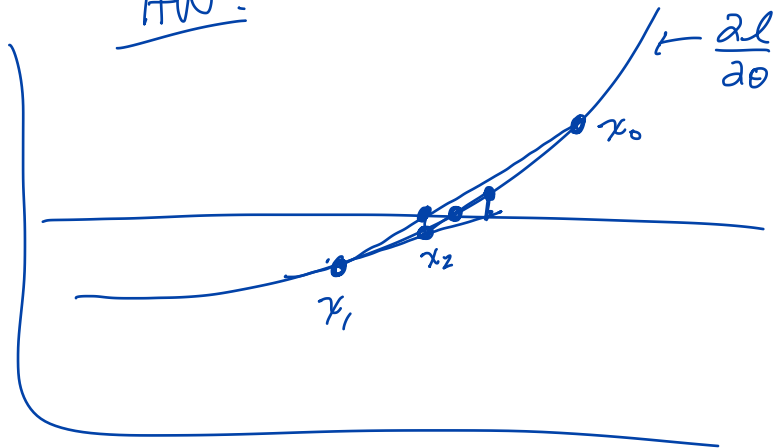
Δ is small

$$\frac{\partial l}{\partial \theta}$$



$$\text{Slope} = \frac{\partial^2 l}{\partial \theta^2}$$

HW.



$l(\theta)$

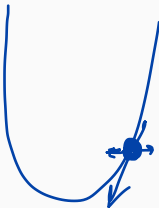
Direct Numeric Optimization

- Based on our last example, we solved for one parameter λ , and numerically found the other, α , via root-finding.
- In many cases, this is not possible. Instead, we might want to directly optimize both parameters.
- We will first introduce this by extending the Newton-Raphson to perform optimization rather than root-finding.

Direct Numeric Optimization II

- The idea is simple: local maximum are just the zeros (roots) of the derivative function.
- Thus, we still perform Newton-Raphson, but on the derivative rather than the original function.
- Starting with initial estimate θ_0 , we update following the equations until convergence:

$$\theta_{n+1} = \theta_n - \frac{f'(\theta_n)}{f''(\theta_n)}$$

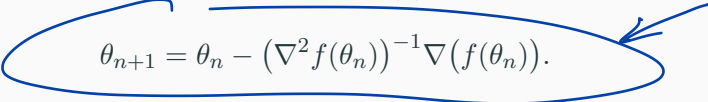


$-l(\theta)$



Direct Numeric Optimization III

- If θ is multivariate, then again we use the **gradient**: $\nabla f(\theta)$, which is a vector, and Hessian: $\nabla^2 f(\theta)$, which is a matrix:


$$\theta_{n+1} = \theta_n - (\nabla^2 f(\theta_n))^{-1} \nabla (f(\theta_n)).$$

- This is the basic approach of many traditional machine learning algorithms:
 - Pick a model that depends on θ , and a loss-function $f(\theta)$ that depends on the data and model (here, our loss is the neg - log-likelihood function).
 - If possible, calculate the derivative of the loss function with respect to θ . If not, approximate numerically.
 - If possible, calculate the Hessian of the loss function with respect to θ . If not, approximate numerically.

Direct Numeric Optimization IV

- Start with initial guess for θ , take steps the size of the (approximate) hessian of $f(\theta)$, in the direction of the gradient $f(\theta)$.
- Many optimization strategies are variants of this basic approach: In practice, Hessians / Gradients may be expensive to compute.
- We will primarily focus our attention on pre-existing solutions.
- In R, the function we will use is called optim.
- There are several optimization methods available within this function

$\text{optim}(\text{par} = \theta_0, \text{fn} = \text{neg-log-like}, \text{gr} = \nabla \text{neg-log-like}, \text{hess} = \text{Hessian}, \text{method})$

Direct Numeric Optimization V

"methods"

$\theta \in \mathbb{R}^{100}$

$\theta \in \mathbb{R}^{1000}$

- Nelder-Mead: a heuristic, direct search method. Does not require differentiability. Slow, but robust.
- BFGS / L-BFGS-B: A Quasi-Newton approach, approximates either (or both) the Hessian and Gradient numerically. Extremely effective for (twice) differentiable functions, but slow as θ grows in dimension.
- SANN: A stochastic approach. Hard to tune, but effective on "rough" surfaces
- CG: Conjugate Gradients. More "fragile" than BFGS, but require less memory storage so can be useful for large θ .
- Brent: Only univariate θ . In these cases, approximates the gradient and hessian, similar to what we proposed. Very effective, but requires univariate θ .

? optim

Direct Numeric Optimization VI

- How it works: Get function f , method you want to use, and starting point θ_0 . Example:

$$\theta = (\alpha, \lambda)$$

```
gamma_negloglik <- function(theta) {  
  -dgamma(  
    x = data,  
    shape = theta[1], rate = theta[2],  
    log = TRUE  
  ) |> sum()  
}
```

$$\left. \begin{array}{l} \text{gamma_negloglik} \\ \text{-dgamma} \\ \text{x = data,} \\ \text{shape = theta[1], rate = theta[2],} \\ \text{log = TRUE} \\ \text{)> sum()} \end{array} \right\} \sum \log L(\theta)$$

Direct Numeric Optimization VII

```
optim(  
  c(2, 5),  
  fn = gamma_negloglik,  
  method = 'L-BFGS-B', # Constrained theta>0  
  lower = c(1e-8, 1e-8) # Constrained theta>0  
)
```

minimize this function

$$\theta = (\alpha, \lambda) \geq 0$$

$\hat{\theta}$ output

[1] 0.9728026 1.9806150

\$value

[1] 6.641716

$fn(\hat{\theta})$

\$counts

function gradient

19

19

gamma-neg log lik(
 (-1, -2)
) $\rightarrow$ error.

Gradient Descent and Variations

- In Newton-Raphson, the Hessian just tells us the size of the step.
- Often the gradient is available, but not the Hessian (or it is expensive to compute the inverse, since it is a matrix).
- In these cases, we often use a different approach called gradient descent.
- Idea: still step in the direction of the (negative) gradient, but the step size will just be small δ_n , and let δ_n shrink over-time:

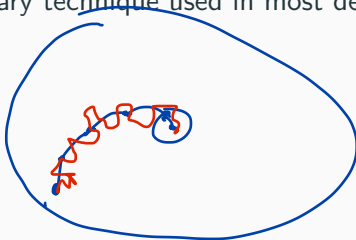
$$\theta_{n+1} = \theta_n - \delta_n \nabla f(\theta_n).$$

Gradient Descent and Variations II

- The idea is simple, but a lot of modern optimization techniques and theory are based on this idea, finding various strategies for specifying δ_n .
- Ex: BFGS approximates the Hessian, but scales poorly with dimension. In deep learning, θ has dimension in millions / billions, so BFGS won't work (nor does optim).
- The success of deep-learning and AI is largely due to the advent of **automatic differentiation**: software that calculates the exact gradient of $f(\theta_n)$, after simple calculations.
- Auto-diff libraries: PyTorch, TensorFlow, JAX, etc. Mostly available in Python, though some are available in R.

Gradient Descent and Variations III

- Final approach we will discuss is stochastic gradient descent, which is effectively randomly sampling the data to compute an approximate gradient.
- The idea is that, even with auto-diff, gradients (with entire data) are expensive to compute.
- Thus, randomly sample data to get a stochastic approximation of the gradient; this is faster to compute, so we get more update-steps.
- This is the primary technique used in most deep-learning frameworks.



Maximum Likelihood: continued examples

Muon Decay

Returning to the Muon-Decay example we used MoM to solve.

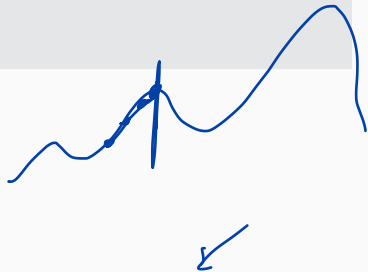
The density of iid observations $X_1, \dots, X_n$ is:

$$f(x; \alpha) = \frac{1 + \alpha x}{2}, \quad -1 \leq x \leq 1 \quad -1 \leq \alpha \leq 1.$$

Find the MLE of α .

$$l(\theta) = l(\alpha)$$

$$= \prod_{i=1}^n \frac{1 + \alpha x_i}{2}$$



$$l(\alpha) = \sum (\log(1 + \alpha x_i)) - n \log(2)$$

$$\frac{\partial \ell}{\partial \alpha} = \sum_{i=1}^n \frac{1}{1 + \exp(x_i)} \cdot x_i = 0$$

$$\sum \frac{x_i}{1 + \exp(x_i)} = 0$$

- $\ell(\alpha)$ into optim ↗ give it gn
 → uniroot($\ell'(\alpha)$) ↘ appox. roots
 → Newton-Raphson. ↘ NM, Brent
- solves for roots of $\ell'(\alpha)$

$d_0 \leftarrow \underline{\text{MoM estimator.}}$

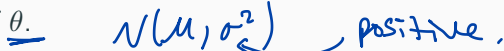
Maximum Likelihood: continued examples II

"If [the likelihood] cannot be maximized analytically, it may be possible to use a computer to maximize [the likelihood] numerically. In fact, this is one of the most important features of the MLE. If a model can be written down, then there is some hope of maximizing it numerically, and, hence, finding MLEs of the parameters." – Casella and Berger (2024, Section 7.2)



Constraint Optimization

Constrained optimization

- In all of our previous examples, we want to maximize the log-likelihood function $f(x_i^*; \theta)$, and there is some natural constraints of θ . 
- Examples: In Gamma(α, λ), $\theta = (\alpha, \lambda)$, and we are constrained by $\alpha, \lambda > 0$.
- In the Gaussian model, $\theta = (\mu, \sigma)$. μ is unconstrained, but $\sigma \geq 0$.
- Similarly, in the Muon-decay example: $\theta = \alpha \in (0, 1)$.
- This leads to the issue of *constrained* optimization, and is typically a requirement for maximum likelihood estimation.
- There are a few common strategies:

Constrained optimization II

- Limit search region to be within constraints: This is the basic approach used by the `uniroot` function. It's also used sometimes in `optim`, e.g., the L-BFGS-B method.
- Variable transformations. Modify the function so that the search algorithm can try all possible values, but apply a transformation that keeps it in the desired range.
 - Example: if $\theta > 0$ is a constraint, optimize over α with $\theta = e^\alpha$. Thus, $\alpha < 0$ is no problem, since $\theta > 0$. (Example on next slide)
 - Another common example is a logit transformation, which ensures $0 \leq \theta \leq 1$, via $\log(\theta/(1 - \theta)) = \alpha$.
- In some cases, the constraint leads to an equation $g(\theta) = 0$, in which case we can apply something like Lagrange-multipliers. We will look at one of these cases as well.

Example: Gamma likelihood

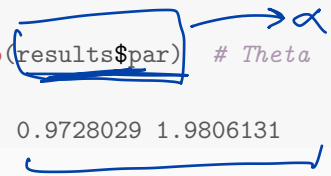
MLE of Gamma likelihood using transformations

Revisit the numeric optimization of the Gamma likelihood, using variable transformations to deal with constraints.

```
gamma_negloglik2 <- function(alpha) {  
  theta <- exp(alpha)  
  -dgamma(  
    x = data,  
    shape = theta[1], rate = theta[2],  
    log = TRUE  
  ) |> sum()  
}
```

Example: Gamma likelihood II

```
results <- optim(  
  c(log(2), log(5)),  
  fn = gamma_negloglik2,  
  method = 'BFGS' # unconstrained  
)  
exp(results$par) # Theta  
  
[1] 0.9728029 1.9806131
```



Edge Color

B,	G,	B,	Other
----	----	----	-------

X_1	X_2	X_3	X_4
-------	-------	-------	-------

$$\sum_{i=1}^4 X_i = N$$

N

P_1, P_2, P_3, P_4

Multinomial Cell probabilities

$$\theta_1^2 + \theta_2^2 = r^2$$

- In this example, we are bound by a curve, not a region. Lagrange Multipliers are a good analytic approach.
- Consider fitting a multinomial distribution to a frequency table.
- We let $X_1, \dots, X_m$ be the counts in cells $1, \dots, m$, and we assume that $(X_1, \dots, X_m)$ follows a multinomial distribution.
- The distribution has parameters n : total counts, and p_i probability of cell i .
- The total count $n = \sum_{i=1}^b X_i$, and p_i being the
- In this case, the X_i are *not* independent, so we don't just take product of IID observations.

Multinomial Cell probabilities II

- However, the likelihood can be calculated via the pmf of the multinomial distribution:

$$L(\theta) = f(x_i^*; \theta) = \frac{n!}{\prod_{i=1}^m x_i!} \prod_{i=1}^m p_i^{x_i^*}.$$

- Given $X_1, \dots, X_m$, the number of trials n is known. The parameter vector of interest is then:

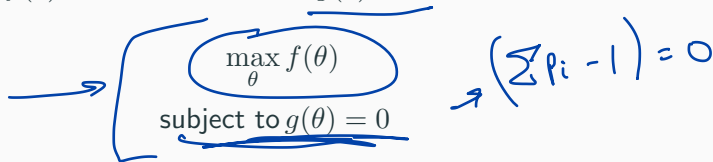
$$\theta = (p_1, \dots, p_m).$$

- The log-likelihood is given by:

$$\ell(\theta) = \log n! - \sum_{i=1}^m \log x_i! + \sum_{i=1}^m x_i \log p_i.$$

Multinomial Cell probabilities III

- Note that we have the constraint $\sum_i p_i = 1$. Thus, we will solve this using the Lagrange-multiplier technique.
- Reminder: Suppose we are trying to maximize (or minimize) a function $f(\theta)$, with the constraint $g(\theta) = 0$. That is,


$$\begin{array}{l} \max_{\theta} f(\theta) \\ \text{subject to } g(\theta) = 0 \end{array} \rightarrow (\sum p_i - 1) = 0$$

- This optimization problem can be solved by introducing the Lagrange function:

$$\mathcal{L}(\theta, \lambda) = f(\theta) + \lambda g(\theta),$$

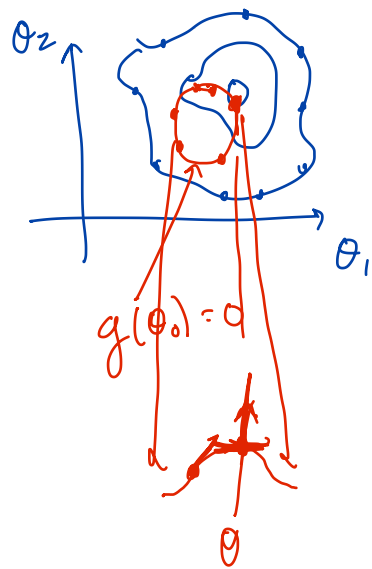
$$g(\theta) = \sum_i p_i - 1 = 0$$

Multinomial Cell probabilities IV

- And solving the system of equations that arises from:

$$\nabla_{\theta, \lambda} \mathcal{L}(\theta, \lambda) = 0.$$

- Recall the partial derivative $\frac{\partial}{\partial \lambda} \mathcal{L}(\theta, \lambda) = g(\theta)$, which must be zero, giving the necessary constraint.
- Then, the condition $\nabla_{\theta} \mathcal{L}(\theta, \lambda) = 0$ ensures that the gradients of f and g are parallel. That is, geometrically, the constraint $g(\theta) = 0$ defines a curve or surface that we are constrained to; on this surface, the max/min value of f occurs when a level-set of f is tangent to the constraint curve $g(\theta) = 0$. This occurs only where $\nabla f(\theta) = \lambda \nabla g(\theta)$, which gives rise to the set of equations defined by $\nabla_{\theta} \mathcal{L}(\theta, \lambda) = 0$.



$$g(\theta) = 0$$

$$\nabla g(\theta) \propto \nabla f(\theta)$$

✓

$$\frac{\partial \mathcal{L}}{\partial \lambda} = g(\theta) = 0$$

✓

Multinomial Cell probabilities V

MLE of multinomial cell probabilities

Use Lagrange multipliers to find the MLE of a multinomial distribution, with $X_1, \dots, X_m$.

$$\rightarrow \ell(\theta) = \log n! - \sum \log x_i^*! + \sum_{i=1}^m x_i^* \log p_i$$

Subject to

$$g(\theta) = \sum_{i=1}^m p_i - 1 = 0$$

$$\mathcal{L}(\theta, \lambda) = \log n! - \sum_{i=1}^m \log x_i^*! + \sum_{i=1}^m x_i^* \log p_i - \lambda (\sum p_i - 1)$$

$$\nabla_{\theta} \mathcal{L}(\theta, x)$$

$$\frac{\partial \mathcal{L}(\theta, x)}{\partial p_j} = \frac{x_j^*}{p_j} - \lambda$$

$$\nabla_{\theta} \mathcal{L}(\theta, x) = \begin{pmatrix} x_1^*/p_1 & -\lambda \\ x_2^*/p_2 & -\lambda \\ \vdots & \\ x_m^*/p_m & -\lambda \end{pmatrix} = 0$$

$$p_i = - \frac{x_i^*}{\lambda} \quad \leftarrow$$

$$\frac{\partial}{\partial \lambda} \mathcal{L}(\theta, \lambda) = g(\theta) = \sum_{i=1}^m p_i - 1 = 0$$

$$= \sum_{i=1}^m p_i = 1$$

$$= - \sum \frac{x_i^*}{\lambda} = 1$$

$$\Rightarrow -\frac{n}{\lambda} = 1$$

$$\Rightarrow \lambda = -n$$

$$\hat{p}_i = \frac{-x_i^*}{\lambda} = \frac{x_i^*}{n}$$

Feb 12, Review

Exam, #2

$$f(k) = p(1-p)^{k-1}$$

$$k = 1, 2, \dots, \dots$$

$$X \sim \text{Geom}(p), \quad E[X] = 1/p$$

$$E[X] = \sum_{k=1}^{\infty} k p (1-p)^{k-1}$$

$$= p \sum_{k=0}^{\infty} k (1-p)^{k-1} = -p \sum_{k=0}^{\infty} \frac{d}{dp} (1-p)^k$$

$$= -p \frac{d}{dp} \sum_{k=0}^{\infty} (1-p)^k$$

$$= -p \cdot \frac{d}{dp} \left(\frac{1}{1-(1-p)} \right)$$

$$= -p \frac{d}{dp} \left(\frac{1}{p} \right)$$

$$= \frac{1}{p}$$

$$\underline{X} \sim \text{Geom}(p)$$

$$X = x^*$$

$$E[X] = \frac{1}{p}$$

$$\overline{\overline{X}} = \frac{1}{p}$$

$$x^* = \frac{1}{p}$$

 $\Rightarrow$

$$\hat{p}_{\text{mom}} = \frac{1}{x^*}$$

$$L(p) = \prod_{i=1}^n f(x^*; p)$$

$$= p(1-p)^{x^*-1}$$



P

Neg-Bin.

Pois(λ)
↑
 p

Pois(λ_{lux})
↕
Pois(λ_{non})

$\left[\frac{\hat{\lambda}_{lux}}{\hat{\lambda}_{non}} \right]$

$\approx \left[\hat{p} \right]$

Multinomial Cell probabilities VI

- Often, we can describe parameters of a model as functions of other parameters.
- Example: in the multinomial model, we have parameters p_i representing probabilities for each cell. We might want to make these a function of θ , such that $p_i(\theta)$; then, we will still want to estimate θ using the data.
- In the case above the log-likelihood is then:

$$\ell(\theta) = \log n! - \sum_{i=1}^m \log x_i^*! + \sum_{i=1}^m x_i^* \log p_i(\theta).$$

Hardy-Weinberg Equilibrium

- Consider alleles expressed as dominant (A) and recessive (a), and they are expressed with rates $1 - \theta$ and θ , respectively.
- The Hardy-Weinberg principle states genotypes are expressed in large, randomly mixing populations following the Punnett square in Table 1.

	A	a
A	$(1 - \theta)^2$	$\theta(1 - \theta)$
a	$\theta(1 - \theta)$	θ^2

Table 1: Genotype frequencies in large, randomly mixing population.

Hardy-Weinberg Equilibrium II

- Since the order of Aa or aA doesn't matter, this results in frequencies for AA , Aa , and aa : $(1 - \theta)^2$, $2\theta(1 - \theta)$, θ^2 .
- Supposing that we observe a population with these three possible gene expressions. Then, using population counts, we want to fit a multinomial model with $m = 3$ categories, and probabilities defined as:

$$p_1 = (1 - \theta)^2, \quad p_2 = 2\theta(1 - \theta), \quad p_3 = \theta^2.$$

Hardy-Weinberg Equilibrium III

Blood Types

In a Chinese population of Hong Kong in 1937, blood types occurred with the following frequencies, where M and N are erythrocyte antigens: M (342), MN (500), N (187), for a total of 1029 samples. Use a multinomial model to estimate the frequency of alleles and genotypes.

$$p(M) = p_m, \quad p(MN) = p_{mn}, \quad p(N) = p_n.$$

$$\rightarrow \underline{p(M) = (1-\theta)^2} \quad \underline{p(MN) = 2\theta(1-\theta)} \quad \underline{p(N) = \theta^2}$$

$$\ell(\theta) = \log n! - \sum_{i=1}^m \log x_i^*! + \sum_{i=1}^m x_i^* \log p_i(\theta)$$

$$\begin{aligned}
 \ell(\theta) = & \log n! - \sum_{i=1}^3 \log x_i^*! + x_1^* \log (1-\theta)^2 - 2x_1^* \log (1-\theta) \\
 & + x_2^* \log 2\theta(1-\theta) \Big] x_2^* (\log 2\theta + \log(1-\theta)) \\
 & + x_3^* \log \theta^2 \Big] x_3^* \cdot 2 \log \theta
 \end{aligned}$$

$$\begin{aligned}
 = & \log n! - \sum_{i=1}^3 \log x_i^*! + (2x_1^* + x_2^*) \log (1-\theta) \\
 & + (2x_3^* + x_2^*) \log \theta \\
 & + x_2^* \log 2
 \end{aligned}$$

$$\frac{\partial \ell(\theta)}{\partial \theta} = - \frac{2x_1^* + x_2^*}{1-\theta} + \frac{2x_3^* + x_2^*}{\theta} = 0$$

$$2x_1^* \theta + \theta x_2^* = 2x_3^* + x_2^* - 2\theta x_3^* - \theta x_2^*$$

$$2\theta x_3^* + 2x_1^* \theta + 2\theta x_2^* = 2x_3^* + x_2^*$$

$$2\hat{\theta} = \frac{2x_3^* + x_2^*}{x_3^* + x_1^* + x_2^*}$$

↓
~~N~~

$$\hat{\theta}_{MLE} = \frac{2x_3^* + x_2^*}{2N}$$

M	MN	N
342	500	187

$$\hat{\theta}_{MLE} = \frac{2(187) + 500}{2(1029)} = 0.4247$$

$$\frac{M}{(1-\theta)^2}$$

$$\frac{MN}{2\theta(1-\theta)}$$

$$\frac{N}{\theta^2}$$

$$\tilde{\theta}^2 \approx \frac{N}{n} = \frac{187}{1029} \quad \text{val}$$

$$\tilde{\theta} \approx \sqrt{\frac{187}{1029}} = 0.4263$$

Invariance property of the MLE

Invariance property

- We will next talk about general properties of estimators, and introduce some theory.
- Before we do that, we still want to discuss point-estimation following the Bayesian approach.
- The order of things is a little tricky, but before moving on we will introduce one property of the MLE that can be used in finding the MLE.
- The property is known as **invariance**.
- Idea: information we have about θ should be the same information we have about $g(\theta)$ – we don't gain or lose information by transforming the variable.

Invariance property II

Theorem: Invariance property of the MLE

(Theorem 7.2.10, Casella and Berger, 2024) If $\hat{\theta}$ is the MLE of θ , then for any function g , $g(\hat{\theta})$ is the MLE of $g(\theta)$. For instance, if $\hat{\theta} = \bar{x}_n^*$, then the MLE of $\sin(\theta)$ is $\sin(\bar{x}_n^*)$.

Invariance property III

- Note in the previous theorem, nothing is preventing θ to be multivariate.
- Thus, if $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_k)$ is the MLE of θ , then $g(\hat{\theta}) = (g(\hat{\theta}_1), \dots, g(\hat{\theta}_k))$ is the MLE of $g(\theta)$.
- For a more concrete example: suppose that p is the probability of an event, a model parameter that we want to estimate. Then if $\hat{p}$ is the MLE, then we can find the MLE for the odds-ratio or log-odds ratio very easily:

$$\text{MLE of odds ratio} = \frac{\hat{p}}{1 - \hat{p}}, \quad \text{MLE of log-odds ratio} = \log \frac{\hat{p}}{1 - \hat{p}}.$$

Comments on maximum likelihood

- There are many advantages of MLE, and is one reason that MLE is one of the gold-standards of statistical estimation (the other dominant approach being Bayesian estimation).
- Maximum likelihood is generally considered a frequentist approach due to asymptotic theory results, which we cover later. As presented, however, the MLE has more of a Bayesian flavor to it: the likelihood measures some degree of “belief” about a parameter, and we estimate the parameter by picking the value that maximizes this belief.
- Maximum likelihood is not without it's criticisms. For instance:

Comments on maximum likelihood II

- The MLE may not be unique (i.e., more than one global maximum). For instance, $\hat{\theta}_1$ and $\hat{\theta}_2$ may both maximize the likelihood $L(\theta)$; if they are very different estimates, this could cause a problem. It's also possible for an infinite number of θ values to maximize the likelihood, which again can be problematic.
- The MLE is often biased, and this can be particularly problematic in small sample sizes.
- The MLE depends on model assumptions. If they are wrong, the estimates might not make much sense.
- The MLE might not even exist! Le Cam, a famous 20th century statistician who was largely a proponent of the MLE approach, wrote a fun paper about some of these weaknesses, and warned against blindly using the MLE (Le Cam, 1990):

Comments on maximum likelihood III

“We present a list of principles leading to the construction of good estimates. The main principle says that one should not believe in principles but study each problem for its own sake.”” – (Le Cam, 1990)

- Other difficulties arise in more complex models, such as time-series or spatial data. Here, the data are not IID, and the distribution of each observation changes over times / space, and observations are not independent. As a result, it's generally very difficult to write down the likelihood, or find the MLE. Minimizing a loss-function is generally easier in complex settings, even if it results in a loss of information (discussed later), or is equivalent to unrealistic Gaussian approximations (i.e., minimizing MSE).

References and Acknowledgements

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References and Acknowledgements II

- Compiled on January 26, 2026 using R version 4.5.2.
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